

SPARTA FORT MCCOY, WI, USA

WMO: 726437

Lat: 43.967N

Lon: 90.733W

Elev: 253

StdP: 98.33

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 94940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.1	(c) -22.3	(d) -29.6	(e) 0.2	(f) -24.4	(g) -26.3	(h) 0.4	(i) -21.2	(j) 8.4	(k) -5.5	(l) 7.6	(m) -6.4	(n) 0.4	(o) 260	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	32.3	22.8	30.9	21.9	28.8	20.8	24.1	29.8	23.1	28.5	22.1	27.0	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.7	27.3	21.3	16.5	26.1	20.4	15.5	25.1	73.9	29.9	69.7	28.3	66.0	27.9	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.3	(c) 6.3	(e) -32.0	(f) 34.3	(g) 3.1	(h) 2.2	(i) -34.2	(j) 35.9	(k) -36.0	(l) 37.2	(m) -37.7	(n) 38.4	(o) -39.9	(p) 40.0		
(a) 8.2	(b) 7.3	(c) 6.3	(e) -32.2	(f) 26.0	(g) 3.0	(h) 1.2	(i) -34.3	(j) 26.9	(k) -36.1	(l) 27.6	(m) -37.8	(n) 28.2	(o) -40.0	(p) 29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-8.2	-7.0	0.3	7.3	13.4	18.7	21.1	19.6	15.6	8.5	1.9	-5.1		
	DBStd	11.60	7.40	6.91	6.89	5.56	4.88	3.71	3.39	3.43	4.67	5.15	5.67	6.41		
	HDD10.0	2302	563	476	310	116	19	0	0	0	7	93	250	468		
	HDD18.3	4304	822	709	559	334	167	38	12	25	108	309	494	726		
	CDD10.0	1299	0	0	9	34	126	263	343	298	174	46	6	1		
	CDD18.3	257	0	0	1	2	14	51	96	65	25	3	0	0		
	CDH23.3	3076	0	0	6	39	211	612	1152	714	304	39	0	0		
	CDH26.7	929	0	0	0	7	55	176	413	202	73	4	0	0		
(14) Wind	WSAvg	2.3	2.3	2.4	2.6	3.1	2.6	2.2	1.9	1.7	1.9	2.3	2.3	2.2		
(15) Precipitation	PrecAvg	889	28	24	47	89	111	132	112	121	86	60	48	31		
	PrecMax	1124	65	46	84	163	275	240	235	380	233	147	135	67		
	PrecMin	629	10	2	8	38	44	46	24	45	16	19	7	11		
	PrecStd	129	14	13	22	37	51	65	51	73	54	34	31	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	12.8	22.9	27.2	31.3	32.8	35.2	33.0	31.4	27.2	20.9	12.5		
		MCWB	5.0	9.7	15.7	16.1	20.7	23.1	23.0	23.6	22.3	19.3	13.5	10.4		
	2%	DB	6.0	7.8	17.7	23.6	27.8	31.1	32.6	31.1	28.7	23.8	16.7	8.0		
		MCWB	3.0	4.8	11.8	14.0	17.9	21.7	23.2	22.1	21.0	16.3	11.6	5.9		
	5%	DB	2.8	4.9	14.1	21.1	25.8	28.7	31.1	28.9	26.9	21.1	13.6	5.5		
		MCWB	1.4	2.5	9.2	12.9	16.9	20.3	22.4	21.2	19.3	15.6	9.4	3.6		
	10%	DB	2.0	2.5	11.1	17.9	23.0	27.0	28.9	27.2	25.0	17.8	11.7	3.0		
		MCWB	0.7	0.6	7.0	10.5	15.7	18.9	21.3	20.0	18.5	13.3	8.1	1.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	8.3	16.5	17.6	22.2	24.7	26.0	24.8	23.5	21.8	14.4	11.4		
		MCDB	8.9	12.9	22.9	24.8	29.3	31.1	31.7	30.2	29.6	25.6	18.1	13.1		
	2%	WB	3.3	5.0	13.5	15.2	20.0	22.8	24.6	23.6	22.0	18.4	12.4	6.2		
		MCDB	5.4	7.5	16.6	22.2	25.4	28.7	30.4	28.5	26.7	22.2	16.0	7.5		
	5%	WB	1.7	2.6	9.6	13.4	18.6	21.6	23.3	22.5	20.7	16.0	10.1	3.6		
		MCDB	3.0	4.7	13.9	19.2	23.7	26.9	29.1	26.9	24.9	20.2	13.4	5.2		
	10%	WB	0.5	0.7	6.8	11.4	16.9	20.3	22.3	21.5	19.4	13.7	7.9	1.9		
		MCDB	1.4	2.1	10.4	17.0	21.6	25.1	27.3	25.7	23.5	17.5	10.7	3.1		
(35) Mean Daily Temperature Range	MDBR		10.4	11.5	11.7	13.1	13.8	13.3	13.8	13.3	13.8	12.3	10.3	9.2		
	5% DB	MCDBR	10.4	12.9	14.9	17.9	16.5	15.4	15.3	14.8	15.3	16.0	14.7	11.3		
		MCWBR	8.3	9.9	8.9	10.0	8.2	7.7	7.2	7.2	8.1	9.6	10.2	9.3		
	5% WB	MCDBR	9.9	12.1	12.5	14.8	13.7	12.9	13.3	12.1	12.9	13.6	13.3	9.9		
		MCWBR	8.1	9.5	8.2	9.1	7.7	7.2	7.2	6.8	7.7	9.3	10.0	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.290	0.325	0.365	0.393	0.404	0.403	0.405	0.365	0.329	0.295	0.265		
	taud		2.400	2.320	2.374	2.333	2.302	2.301	2.314	2.309	2.411	2.502	2.516	2.458		
	Ebn at Noon		879	903	914	898	879	867	863	848	859	852	834	841		
	Edh at Noon		74	97	105	120	127	128	125	121	101	81	67	64		
(44) All-Sky Solar Radiation	RadAvg		1.77	2.87	3.91	4.51	5.37	5.94	6.14	5.22	4.19	2.68	1.74	1.30		
	RadStd		0.20	0.26	0.47	0.43	0.41	0.46	0.30	0.35	0.32	0.34	0.15	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page